

Computer Engineering Department

King Mongkut's University of Technology Thonburi

CPE 241 Database

Individual Lab #02 Modified Web App

Date: Check Individual 02 assessment activities on LEB2

Introduction

In Individual 01, you successfully installed and ran the reference web application **InvoiceLab** in your local environment. This allowed you to observe how the front-end, back-end, and database interact as a complete system.

In this lab, you will begin modifying the existing application to add new components. The purpose of this exercise is to help you understand how changes in the database schema, APIs, and user interface must work together across the technology stack.

You will extend the system by introducing a new entity called **Sales Person** and integrating it with the existing **Invoice** module. This activity will demonstrate how relational data models are connected and how those relationships propagate through backend services and user interface components.

Objectives

1. To understand how modifications to a web application affect multiple layers of the system, including the database, backend APIs, and user interface.
2. To design and implement a new relational entity (Sales Person) and integrate it with the existing Invoice module using a foreign key relationship.
3. To modify backend APIs and database schemas so that the new entity can be stored, retrieved, and associated with invoice data.
4. To update user interface components to display, select, and print sales person information within the invoice workflow.

Instructions

STEP 1: Prepare needed material and learn from them

- Download the InvoiceLab app from GitHub.
<https://github.com/llbumpbumpl/InvoiceDoc2>
- Check LEB2 -> Assessment Actitiy -> Individual 02 – Modified Web App. Follow the system instructions video link to learn about system working before beginning the lab.
https://drive.google.com/drive/folders/1YTbQq5fNkVbuo6KUhPM_mDUPVWh-auhO?usp=drive_link
- After watching the videos, follow the lab instructions from this link.
<https://github.com/llbumpbumpl/InvoiceDoc2/tree/main/labs/lab08>
The PDF file of the required slides can also be found there.

STEP 2: Modify the InvoiceLab app

The modified app must have the following features:

- Add a new model in the database only: sales_person(id, code, name, start_work_date).
You need not yet add the forms. Populate this with at least 10 rows of data.
- Add a sales person to the invoice header which will have behavior similar to the customer.
 - In invoice header table invoice add field invoice.sales_person_id which is a foreign key of sales_person.id. This is not a required field.
 - Modify the invoice APIs to include the sales person information.
 - In the UI, the invoice *list view* should also show the sales_person.name.
 - In the UI, the invoice *form view* should also show sales person code (user can key), [List of Value] button, and sales person name (cannot edit).
 - Add a sales person *list view* for when the [List of Value] button is pressed.
 - Modify the printed invoice to also show the sales person code and name.

Deliverables

Submit a PDF to show that you have successfully accomplished the above tasks by:

1. Show a screen capture of your new invoice list view that also shows the name of the **sales person** (could be null). At least 3 of your invoices should have a **sales person** (you can create a new invoice with a sales person or edit an existing one to add a sales person). The **sales person** in remaining invoices can be null.
 2. Show a screen capture of your revised invoice form after a new invoice with a **sales person** is created (saved).
 3. Show a screen capture of a printed invoice that has a **sales person**.
 4. Show a screen capture of when the [List of Value] button to select a **sales person** is pressed. This will show the **sales person** list view.
 5. Show the source code for all the APIs you modified. Do not include those you did not modify.
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Final Submission

- Submit a PDF file and ensure it is named appropriately (e.g., **IND02_YourStudentID.pdf**) then submit the report on LEB2 before the deadline.
- Upload the PDF report to the Assessment Activities section for Individual02 on LEB2.